

Module 4 Question Pool

OMSCS 6250 Computer Networks

Lesson 4: AS Relationships and Interdomain Routing

Autonomous Systems and Internet Interconnection

Q1. [MCQ]

Consider a network of access ISPs, regional ISPs, Tier-1 ISPs, IXPs and a content provider. Access ISP A knows how to reach only its own customers. It does not know a path to any other network. A host in access ISP A sends a packet to a distant network.

Which network does access ISP A rely on to carry the packet toward the destination?

- A. A regional or Tier-1 ISP that A pays for transit, which forwards it on.
- B. An access ISP that A peers with, which passes it on to its own providers.
- C. An IXP where A is a member, which routes it toward the destination using its own routing table.
- D. A content provider that A connects to, which relays it to other networks.

Correct answer: **A**

Why: Access ISP A knows routes only to its own customers. For everything else it relies on a provider, a regional or Tier-1 ISP that it pays for transit, and that provider forwards the packet toward the destination. An IXP switches frames between the routers of its members. It does not route packets, so it does not carry traffic on to other networks. A peer of A carries traffic between its own customers and those of A, but it will not pass the traffic of A on to its providers. A content provider connects to other networks to deliver its own content, not to relay traffic for others.

Q2. [MCQ]

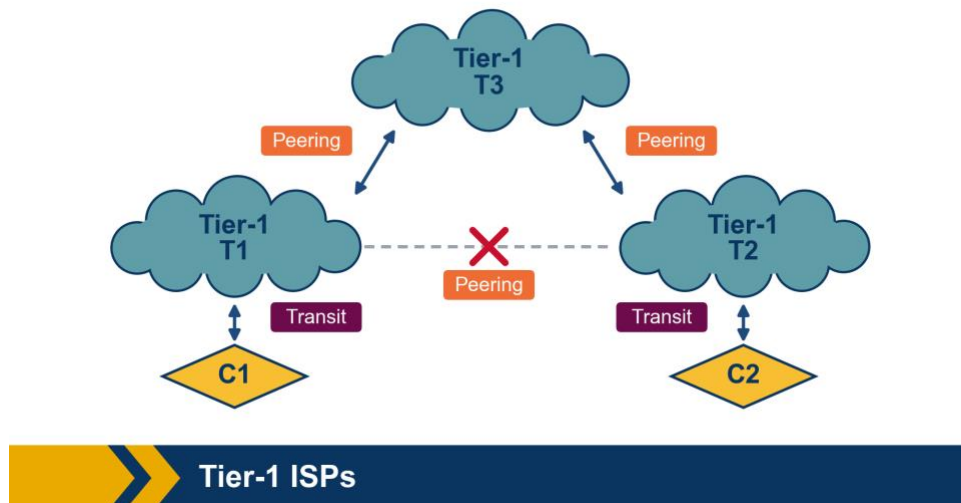


Figure: Tier-1 ISPs (Module 4)

Consider three Tier-1 ISPs, T1, T2 and T3. Each pair of Tier-1 ISPs has a direct peering link. None of the three Tier-1 ISPs has a provider. AS C1 is a customer of T1 only. AS C2 is a customer of T2 only. Every AS exports routes according to its business relationships. The peering link between T1 and T2 fails and stays down.

What happens to the traffic that C1 sends to C2?

- A. It reaches C2 through T3, which peers with both T1 and T2.
- B. It reaches C2 through a default route that T1 keeps for failures.
- C. It is dropped, since T1 has no route to C2 and no default route.
- D. It is dropped until T3 starts passing C2's route on to T1.

Correct answer: C

Why: After the link fails, the only physical path left goes from C1 to T1, then to T3, then to T2, and then to C2. T3 learns the route to C2 from its peer T2. An AS advertises routes learned from a peer only to its customers, so T3 does not advertise the route to C2 to its peer T1. Neither T1 nor T2 pays T3, so carrying traffic between them would earn T3 nothing. As a result, T1 has no route to C2. T1 has no provider, so it also has no default route. The traffic from C1 to C2 is dropped. T3 will not start advertising the route unless it changes its export policy.

Q3. [MCQ]

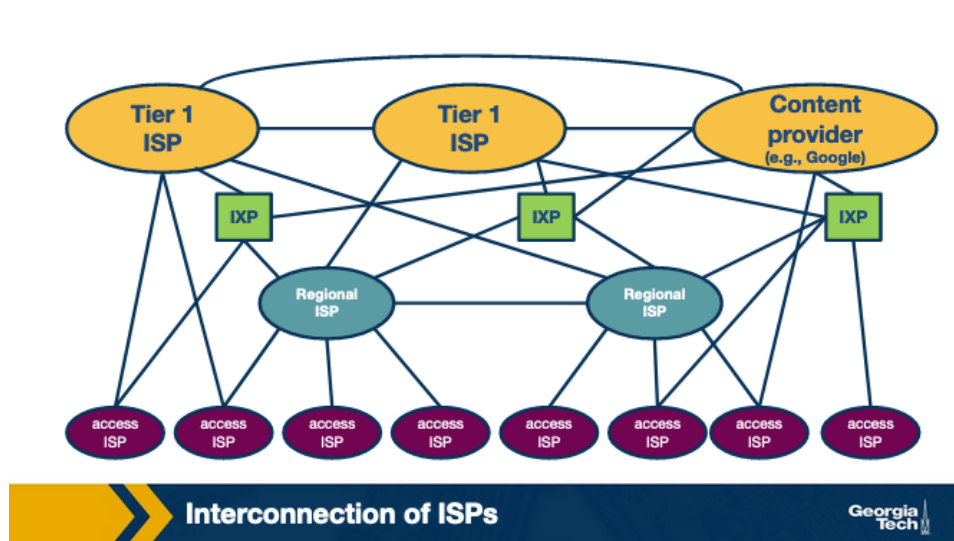


Figure: Interconnection of ISPs, IXPs and a content provider (Module 4)

Consider a network of access ISPs, regional ISPs, Tier-1 ISPs, IXPs and a content provider. In the early Internet, most traffic between networks traveled up the hierarchy, from access ISPs through regional ISPs to Tier-1 ISPs. Today the structure of the Internet is flatter.

Which development drove most of this flattening?

- A. Tier-1 ISPs began to peer with each other instead of buying transit.
- B. Regional ISPs now buy transit from CDNs instead of from Tier-1 ISPs.

Q5. [MCQ]

ISP X and ISP P have a customer-provider relationship. ISP X is the customer. ISP P is the provider.

What does ISP P provide to ISP X under this relationship?

- **A.** ISP P forwards the traffic of ISP X to destinations P has routes to.
- **B.** ISP P forwards traffic sent by ISP X, but not traffic sent toward X.
- **C.** ISP P forwards the traffic of ISP X only to the customers of ISP P.
- **D.** ISP P forwards the traffic of ISP X while the traffic ratio is balanced.

Correct answer: **A**

Why: A provider forwards its customer's traffic to every destination it has a route to, and it carries the traffic sent back to the customer as well. Forwarding traffic only to the provider's own customers is what a peer does. A balanced traffic ratio matters in a peering agreement. Between a customer and its provider, the customer pays whatever the ratio.

Q6. [MCQ]

ISP A and ISP B are of similar size. Each of the two ISPs buys transit from its own providers. Much of the traffic from ISP A goes to the customers of ISP B. Much of the traffic from ISP B goes to the customers of ISP A. ISP A and ISP B set up a settlement-free peering link between them.

What is the main motivation for this peering link?

- **A.** To keep the traffic exchanged between them balanced in each direction.
- **B.** To give each other's customers transit to the rest of the Internet.
- **C.** To reach each other's peers as well as each other's customers.
- **D.** To stop paying providers for traffic sent to each other's customers.

Correct answer: **D**

Why: Both ISPs pay their providers for transit, including for the traffic they send to each other's customers. A direct peering link carries that traffic without payment to a provider. Under the usual export rules, each ISP advertises to its peer the routes to its own customers, and it passes routes learned from its providers and other peers only to its customers. Peering therefore gives neither ISP transit to the rest of the Internet, nor a path to the peers of the other ISP. A balanced traffic ratio helps a peering agreement last, but it is not why two ISPs start peering.

Q7. [MCQ]

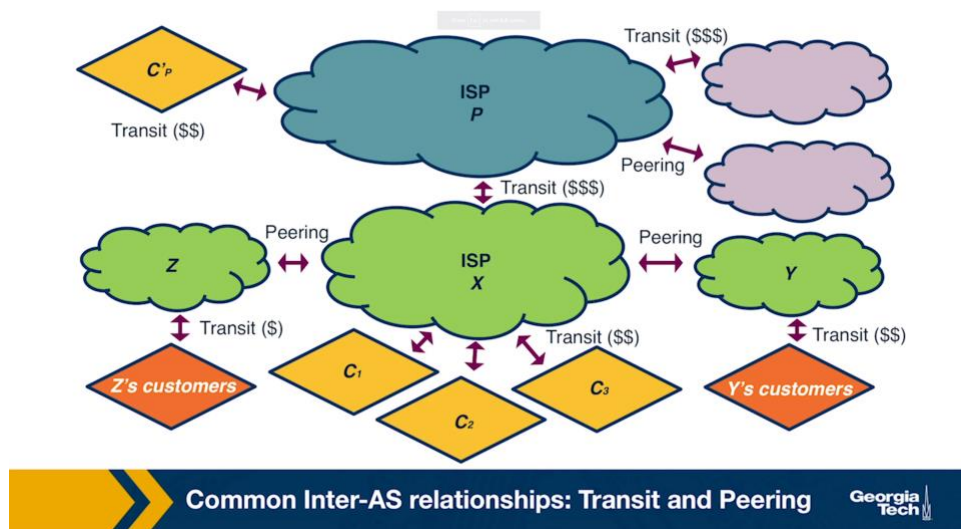


Figure: Transit and peering relationships (Module 4)

Consider AS X, which has customers C1, C2 and C3, peers Y and Z, and provider P. AS X sets LocalPref according to its business relationship with each neighbor. AS X learns four routes to the same prefix p. The four routes come from customer C1, peer Y, peer Z and provider P. The route from provider P has the shortest AS path of the four. The route from peer Y carries the lowest MED of the four. Router R in AS X compares the four routes. Router R has the lowest IGP cost to the border router that learned the route from peer Z.

Which route does router R select?

- A. The route from customer C1.
- B. The route from provider P.
- C. The route from peer Z.
- D. The route from peer Y.

Correct answer: **A**

Why: AS X gives customer routes the highest LocalPref, peer routes the next highest, and provider routes the lowest. LocalPref is the first step of the decision process, and router R looks at later steps only when LocalPref ties. The route from customer C1 wins at that first step. The shorter AS path of provider P, the lower MED of peer Y and the lower IGP cost toward peer Z would matter only in a tie, at steps 2, 4 and 6.

Q8. [MCQ]

AS X has customers, peers and providers. AS X learns routes to the same destinations from all three kinds of neighbor.

Why does AS X prefer routes learned from customers?

- A. Traffic sent to a customer earns AS X revenue, since the customer pays X.
- B. Customer routes have the fewest AS hops, so they give the lowest delay.
- C. Peer routes are billed by traffic volume, while customer routes cost X nothing.
- D. BGP path selection ranks customer routes first, before it compares LocalPref.

Correct answer: **A**

Why: A customer pays AS X, and the more traffic AS X carries for it, the more AS X earns. Sending traffic over a customer route brings in money. BGP itself knows nothing about business relationships. The operator sets a higher LocalPref on customer routes, and BGP compares LocalPref first. Peer routes are usually free under the peering agreement rather than billed by volume, and AS path length counts AS hops, not delay.

Q9. [TF]

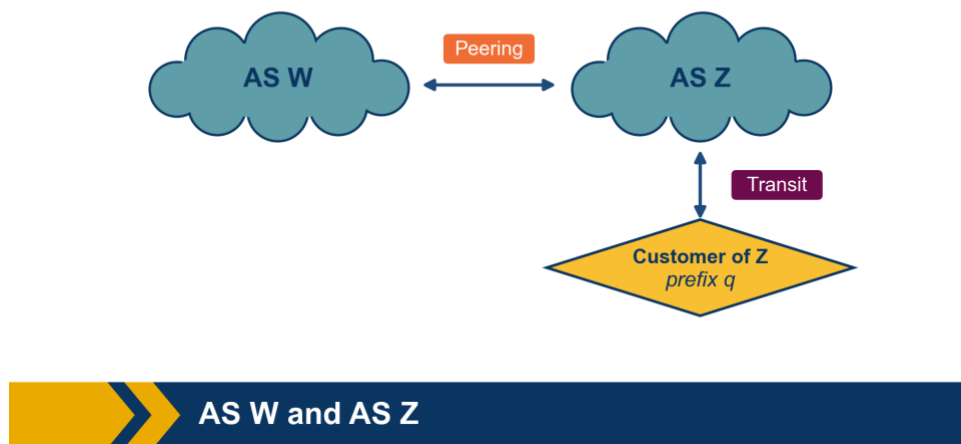


Figure: AS W and AS Z (Module 4)

Consider AS Z and its peer, AS W. Prefix q belongs to a customer of AS Z.

AS W sends traffic toward q over its link to AS Z only after AS Z advertises the route to q to AS W.

- True
- False

Correct answer: **True**

Why: Traffic flows in the opposite direction to route advertisements. AS W sends traffic toward q over its link to AS Z only if it has a route to q over that link, and it gets that route only when AS Z advertises it. In this way, an AS controls the traffic entering its network through the routes it advertises.

Q10. [TF]

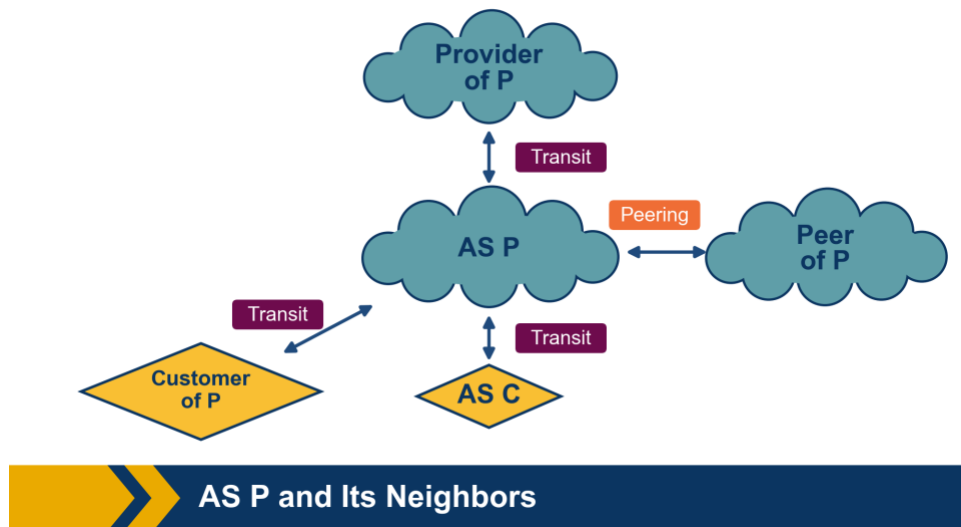


Figure: AS P and its neighbors (Module 4)

Consider AS P and its neighbors. AS C buys transit from AS P. The routing table of AS P holds routes that P learned from its customers, from its peers and from its providers. AS P exports routes according to its business relationships.

AS C can reach every destination in the routing table of AS P through P.

- True
- False

Correct answer: **True**

Why: A provider advertises all of its routes to its customers, whether it learned them from customers, peers or providers. As a result, AS C learns a route through AS P to every destination in P's routing table. That reach is what AS C pays AS P for.

Q11. [TF]

Provider P charges its customer C based on bandwidth use. P measures the bandwidth that C uses every five minutes over a 30-day month. P then charges C by the 95th percentile of these measurements. The traffic of C is very high for 20 minutes. For the rest of the month, the traffic of C stays at the same low rate.

The 20 minutes of high traffic increase the amount that P charges C.

- True
- False

Correct answer: **False**

Why: A 30-day month has 8,640 five-minute measurements. The highest 5 percent of them, 432 measurements, are above the 95th percentile, so they do not set the bill. The 20 minutes of high traffic give only 4 measurements. These 4 are among the highest 432, so the bill stays at the low rate.

BGP Routing Policies: Importing and Exporting Routes

Q12. [MCQ]

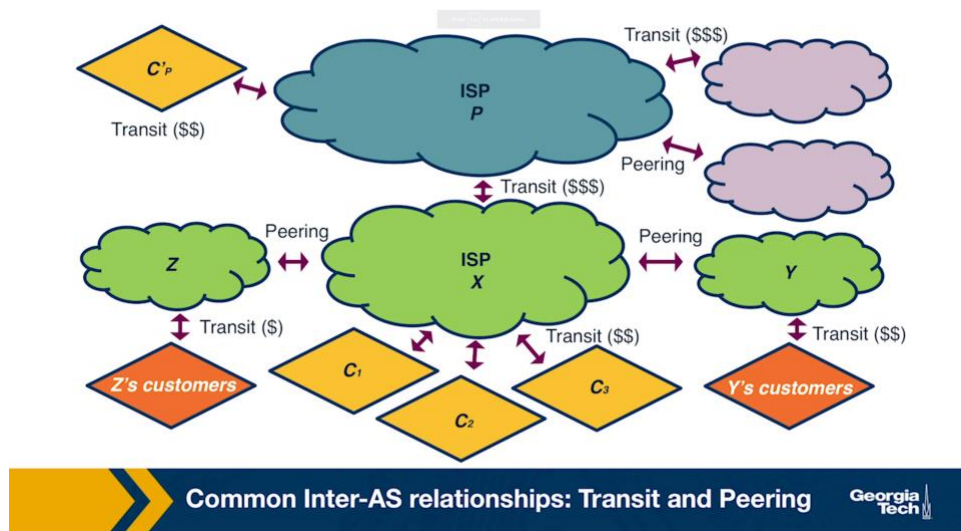


Figure: Transit and peering relationships (Module 4)

Consider AS X, which has customers, peers and providers. AS X exports routes according to its business relationships. AS X learns a route to prefix p from one of its customers.

To which neighbors does AS X advertise this route?

- A. To its providers and its peers, not its customers.
- B. To its customers, its peers and its providers.
- C. To its customers only, and to no peer or provider.
- D. To its customers and its peers, not its providers.

Correct answer: **B**

Why: The customer pays AS X to make it reachable, and the more traffic reaches the customer through AS X, the more AS X earns. AS X therefore advertises the customer's route to all of its neighbors: customers, peers and providers. Keeping the route from its peers or providers would turn away traffic that AS X is paid to carry.

Q13. [MCQ]

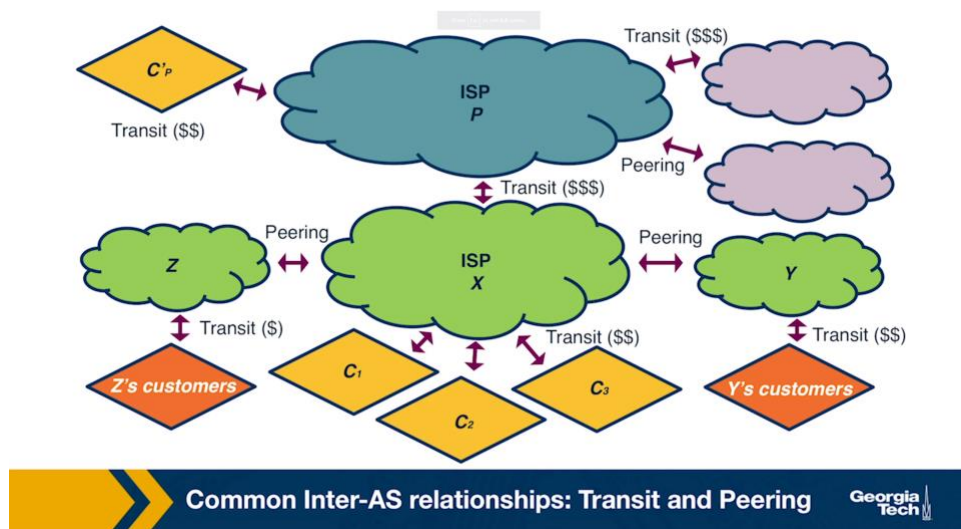


Figure: Transit and peering relationships (Module 4)

Consider AS X, which has customers, peers and providers. AS X exports routes according to its business relationships. AS X learns a route to prefix p from one of its providers.

To which neighbors does AS X advertise this route?

- A. To its customers and its peers, not its providers.
- B. To its customers, its peers and its providers.
- C. To its customers only, and to no peer or provider.
- D. To its providers and its peers, not its customers.

Correct answer: C

Why: AS X advertises a route learned from a provider only to its customers. The customers pay AS X, so carrying their traffic over this route earns AS X money. If AS X advertised the route to a peer or to another provider, those networks would send traffic over it. AS X would pay its provider to carry that traffic and earn nothing from it.

Q14. [MCQ]

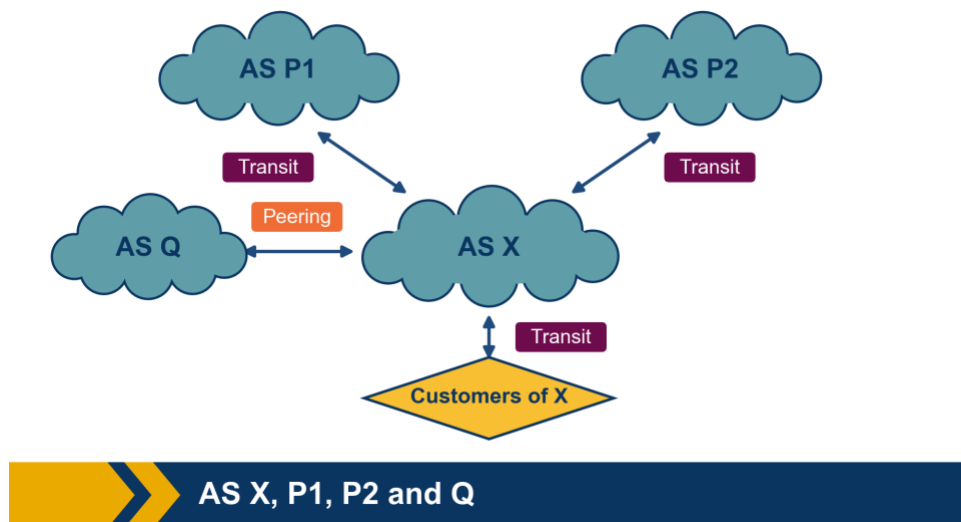


Figure: AS X, P1, P2 and Q (Module 4)

Consider AS X and its neighbors. AS X buys transit from two providers, P1 and P2. AS X also has a peering relationship with AS Q. AS X learns a route to prefix p from P1. AS X advertises this route to its customers, but not to P2 or Q.

Why does AS X not advertise the route to P2 and Q?

- A. BGP rejects a route passed from one provider to another, to stop loops.
- B. P2 and Q hear the route from P1 anyway, so X's copy is redundant.
- C. X would carry their traffic to P1 and pay P1 for it, earning nothing.
- D. X cannot pass on the MED that P1 attached to its route to p.

Correct answer: C

Why: Traffic flows in the opposite direction to route advertisements. If AS X advertised the route to P2 or Q, they would send their traffic for p to AS X, which would then pay P1 to carry it and get nothing in return. That cost stays the same even if P2 and Q also learn the route from P1. MED is not passed beyond the next AS, but that limits only the MED value, not whether AS X advertises the route. The BGP loop check is not the reason either, since it discards only a route whose AS path already contains the receiver's own AS number.

BGP and Design Goals

Q15. [MCQ]

Inside an AS, an IGP such as OSPF optimizes a path metric under one administrative authority. Between ASes, BGP attaches a path vector to each route. The path vector is the list of ASes that the route advertisement has crossed.

Which property of this design made BGP suitable for routing between ASes?

- A. Each route follows the fewest AS hops, which gives the lowest delay.
- B. The AS list proves which AS owns a prefix, so false routes are rejected.
- C. Each router gets a full map of every AS's internal links to compute best paths.

- **D.** Each AS keeps its policy private, and the AS list still blocks loops.

Correct answer: **D**

Why: BGP lets each AS choose which routes to import and export and keep those choices to itself. The list of ASes in each route still prevents loops, because a router discards any route whose AS list already contains its own AS number. A full map of every AS's internal links is how a link-state IGP works, and competing ASes do not share their internal links. BGP never checks who owns a prefix, so the AS list proves nothing about ownership. The number of AS hops is not a measure of delay, and BGP compares LocalPref before AS path length.

BGP Protocol Basics

Q16. [MCQ]

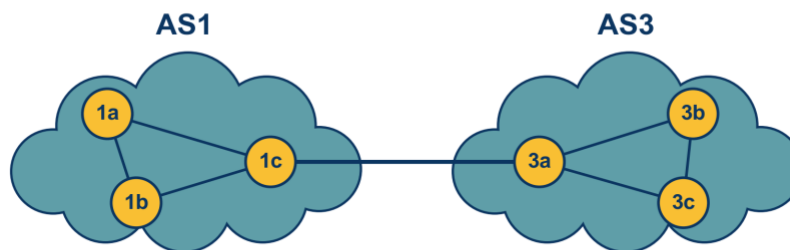


Figure: AS1 and AS3 (Module 4)

Consider two neighboring ASes, AS1 and AS3. Router 1c is in AS1, and router 3a is in AS3. Routers 1c and 3a are connected by a direct link and exchange routing information over it.

What kind of session do routers 1c and 3a run?

- **A.** An IGP session, since BGP runs only inside one AS.
- **B.** eBGP, because 3a and 1c belong to different ASes.
- **C.** iBGP, because 3a and 1c are both border routers.
- **D.** iBGP, because 3a and 1c share a direct link.

Correct answer: **B**

Why: Routers 1c and 3a are in different ASes, which makes their session external BGP (eBGP). iBGP runs between two routers in the same AS, whether or not they are border routers. A direct link does not point to iBGP either, since eBGP sessions usually run over one. Between ASes, routers exchange routes with BGP, not with an IGP.

Q17. [MCQ]

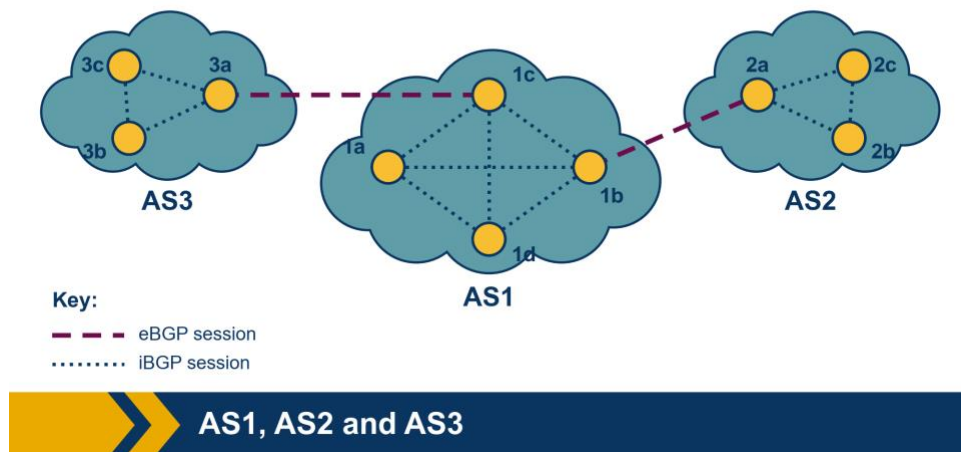


Figure: AS1, AS2 and AS3 (Module 4)

Consider three ASes, AS1, AS2 and AS3. Router 3a in AS3 has an eBGP session with router 1c in AS1. Router 3a has learned routes to several prefixes from router 1c and holds them in its routing table. Router 3a has no other route to these prefixes. Router 3a stops receiving KEEPALIVE messages from router 1c. No message arrives within the maximum time that router 3a waits before it gives up on the session.

What does router 3a do with the routes it learned from router 1c?

- A. Router 3a keeps the routes until 1c announces them again.
- B. Router 3a keeps the routes until 1c sends a withdrawal message for each one.
- C. Router 3a deletes the routes and sends a withdrawal message to its other neighbors.
- D. Router 3a deletes the routes and sends no withdrawal message to any neighbor.

Correct answer: C

Why: Router 3a hears nothing from router 1c within its waiting time, so it closes the session. The routes it learned over that session can no longer be used, and it deletes them from its routing table. It also sends a withdrawal message to its other neighbors, because otherwise they would keep sending traffic over routes that no longer work. While the session is down, router 1c cannot send any withdrawal or new announcement to router 3a.

Q18. [MCQ]

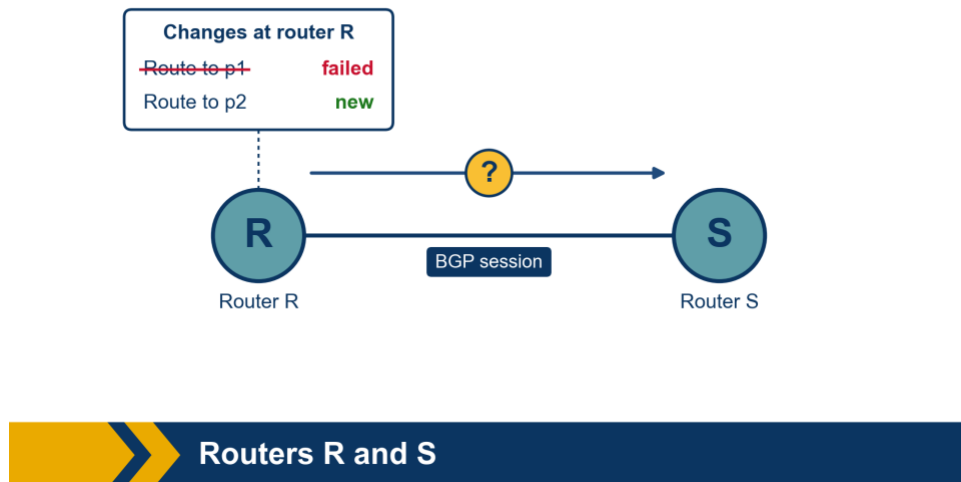


Figure: Routers R and S (Module 4)

Consider routers R and S, which have a BGP session. Router R previously announced a route to prefix p1 to router S. That route has now failed. Router R has also learned a new route to prefix p2 since its last message to router S. Router R must inform router S of both changes.

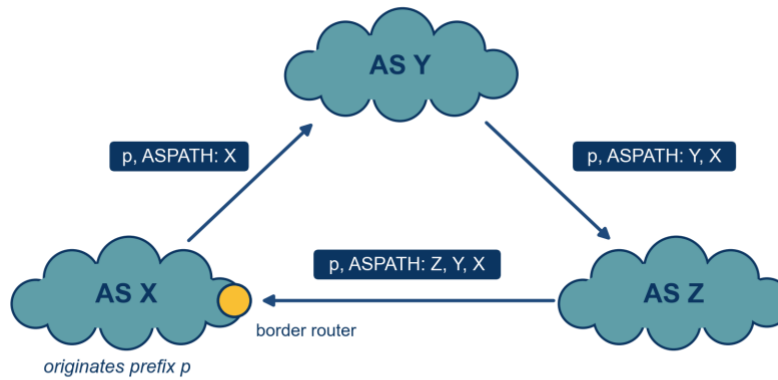
Which BGP message or messages does router R send to reflect these changes?

- A. UPDATE for the new route, and a separate WITHDRAW for the other.
- B. UPDATE, which carries both announcements and withdrawals.
- C. OPEN, which restarts the session and resends the full routing table.
- D. KEEPALIVE, which reports the routes that changed since the last one.

Correct answer: **B**

Why: BGP reports changes in UPDATE messages, and one UPDATE can both announce new routes and withdraw routes that are no longer available. There is no separate WITHDRAW message in BGP. A KEEPALIVE only keeps the session open when there is nothing else to send. An OPEN message starts a new session and carries no routes. On a session that is already open, like the one between R and S, BGP sends only the changes.

Q19. [TF]



AS X, AS Y and AS Z

Figure: AS X, AS Y and AS Z (Module 4)

Consider three ASes, AS X, AS Y and AS Z. AS X originates prefix p and advertises it to AS Y. AS Y advertises the route to AS Z. AS Z then advertises the route back to a border router of AS X. The ASPATH of this announcement lists AS Z, AS Y and AS X.

The border router of AS X discards the announcement.

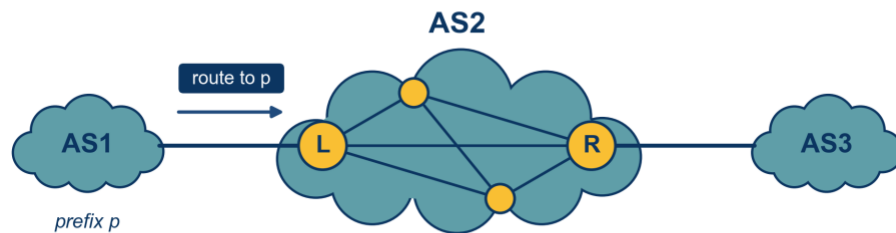
- True
- False

Correct answer: **True**

Why: When a route arrives from another AS, the border router looks for its own AS number in the ASPATH. AS X is already there, and using the route would send packets around in a loop. To prevent that, the border router of AS X discards the announcement.

iBGP and eBGP

Q20. [MCQ]



AS1, AS2 and AS3

Figure: AS1, AS2 and AS3 (Module 4)

Consider three ASes, AS1, AS2 and AS3, connected in a line. AS2 has two border routers, L and R. Router L connects to AS1, and router R connects to AS3. Router L learns a route to prefix p from AS1 over eBGP.

How does router R learn the route to prefix p?

- A. From AS2's IGP, which carries p with its ASPATH attached.
- B. From L, over the iBGP session that L and R hold inside AS2.
- C. From AS3, which learns p from AS2 and advertises it back.
- D. From AS1 directly, over an eBGP session of its own.

Correct answer: **B**

Why: Border routers learn routes from other ASes over eBGP and pass them to the other routers in their own AS over iBGP. In AS2, router L passes the route to router R over their iBGP session. Router R has no session with AS1. If AS3 sent the route back, its ASPATH would already contain AS2, and router R would discard it. An IGP does not carry BGP attributes, so the ASPATH would be lost.

Q21. [MCQ]

Consider an AS with 5 BGP routers, connected in a full mesh of iBGP sessions. The operator adds a sixth BGP router to the AS and keeps the full mesh.

How many new iBGP sessions does the operator need to set up?

- A. 0, since the new router learns BGP routes from the IGP.
- B. 15, the number of sessions in a full mesh of 6 routers.
- C. 1, from the new router to its nearest existing router.
- D. 5, one from the new router to each existing router.

Correct answer: **D**

Why: In a full mesh, every BGP router has an iBGP session with every other BGP router. The new router therefore needs one session to each of the 5 existing routers, which makes 5 new sessions. The mesh grows from 10 sessions to 15, so 15 is the new total rather

than the number added. A single session to one router is how a route reflector client connects, and an IGP does not carry BGP routes with their attributes.

Q22. [TF]

Consider AS X with routers R1 and R2. The two routers hold an iBGP session, but they are not directly connected.

The iBGP messages between R1 and R2 travel along paths that the IGP of AS X computes.

- True
- False

Correct answer: **True**

Why: iBGP runs on top of the IGP. The IGP computes the paths between the routers inside AS X, and the iBGP messages between R1 and R2 follow those paths. That is why the two routers need no direct link to hold an iBGP session.

BGP Decision Process: Selecting Routes at a Router

Q23. [MCQ]

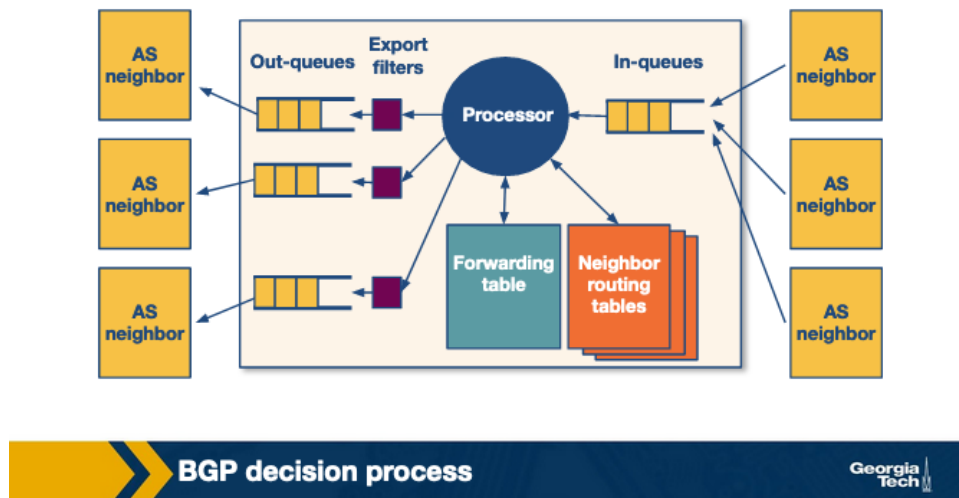


Figure: BGP route-handling pipeline (Module 4)

Consider a BGP router that receives route advertisements from its neighbors.

In what order does the router process each route advertisement?

- A. Export policy, decision process, forwarding table, import policy.
- B. Decision process, import policy, forwarding table, export policy.
- C. Import policy, forwarding table, decision process, export policy.
- D. Import policy, decision process, forwarding table, export policy.

Correct answer: **D**

Why: The router first applies its import policy, which removes the routes it will not consider. It then runs the decision process to select the best route. It installs the best route in the forwarding table. Last, it applies its export policy to decide which neighbors receive the route. If the decision process ran before the import policy, it could select a route that the policy removes.

Q24. [MCQ]

Router X has three eBGP routes to prefix p:

Route 1: LocalPref 100, an AS path of 3 ASes, MED 10

Route 2: LocalPref 100, an AS path of 2 ASes, MED 50

Route 3: LocalPref 80, an AS path of 1 AS, MED 0

Which route does router X select?

- **A.** Route 2, since it ties Route 1 on LocalPref and has a shorter path.
- **B.** Route 1, since it ties Route 2 on LocalPref and has the lower MED.
- **C.** Routes 1 and 2, since they tie on LocalPref and X splits the traffic.
- **D.** Route 3, since it has the lowest LocalPref and the shortest path.

Correct answer: **A**

Why: The router first keeps the routes with the highest LocalPref. Route 3, at LocalPref 80, drops out despite its one-AS path. Routes 1 and 2 tie at LocalPref 100, and the router compares AS path length next. Route 2 wins with a path of 2 ASes against 3. MED comes later in the order and is never reached. Each later step only breaks a tie, so the router always ends with one route.

Q25. [MCQ]

Consider AS B and its two neighbors, AS A and AS C. AS B learns routes to the same prefix from AS A and from AS C. The route from AS C has a shorter AS path than the route from AS A. AS B wants its outbound traffic to that prefix to leave through AS A.

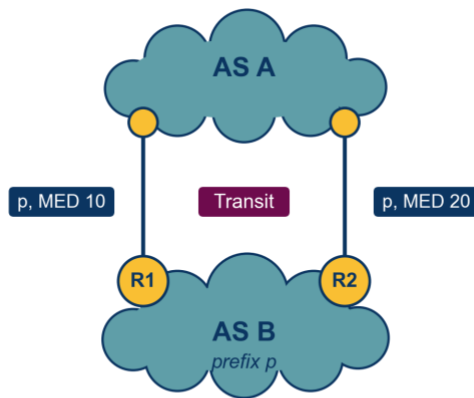
What does AS B configure to achieve this?

- **A.** A lower MED on the routes B advertises to A than on those to C.
- **B.** Extra copies of B's AS number on the routes B advertises to C.
- **C.** A higher LocalPref on the route from A than on the route from C.
- **D.** Lower IGP costs to the border routers that connect B to A.

Correct answer: **C**

Why: AS B sets LocalPref on the routes it learns, and BGP compares LocalPref first. A higher LocalPref on the route from AS A makes the routers of AS B send the traffic out through AS A, even though the route from AS C has a shorter AS path. MED and AS path prepending change the routes that AS B advertises to others. They affect the traffic coming into AS B, not the traffic leaving it. IGP cost is compared after AS path length, so lower IGP costs toward AS A cannot beat the shorter path from AS C.

Q26. [MCQ]



AS A and AS B

Figure: AS A and AS B (Module 4)

Consider AS B, which connects to AS A by two links. One link ends at router R1 in AS B, and the other link ends at router R2 in AS B. AS A is the provider of AS B and honors the MED values that AS B sets. AS B advertises prefix p on both links, with MED 10 on the link at R1 and MED 20 on the link at R2. At AS A, the two routes to p tie on LocalPref, AS path and origin type.

Where does the traffic from AS A to prefix p enter AS B?

- A. Through the link to whichever of R1 and R2 has the lower router ID.
- B. Through R1, since A prefers the lower MED among otherwise tied routes.
- C. Over both links, since A splits its traffic between the two routes.
- D. Through the link closer to A's sending router by A's IGP cost.

Correct answer: **B**

Why: At AS A, the two routes to p tie on LocalPref, AS path and origin type, so AS A compares their MED values next and picks the lower one. The route through R1, with MED 10, wins, and the traffic enters AS B at R1. The decision process ends with a single route, which means AS A does not split the traffic. It would compare IGP costs only if the MED values were equal, and router IDs only after that.

Q27. [MCQ]

Consider two neighboring ASes, AS A and AS B, connected by two links. AS B advertises prefix p to AS A over both links, with a different MED value on each link.

Which exit choices do these MED values affect?

- A. B's exits toward A, since B picks which link its own traffic leaves on.
- B. A's exits toward B, since A picks which link its traffic leaves on.
- C. The exits of A's neighbors, since A passes B's MED values on with the route.
- D. A's exits toward other ASes, since A compares B's MED with their routes.

Correct answer: **B**

Why: Without MED, AS A sends traffic to AS B over whichever link is cheapest for AS A. MED lets AS B ask AS A to use a different link, but AS A makes the choice, over its own links toward AS B. In effect, MED shapes the traffic that enters AS B. AS A does not pass the MED on to other ASes, and it compares MED values only between routes from the same neighbor AS. AS B chooses its own exits toward AS A with LocalPref.

Q28. [MCQ]

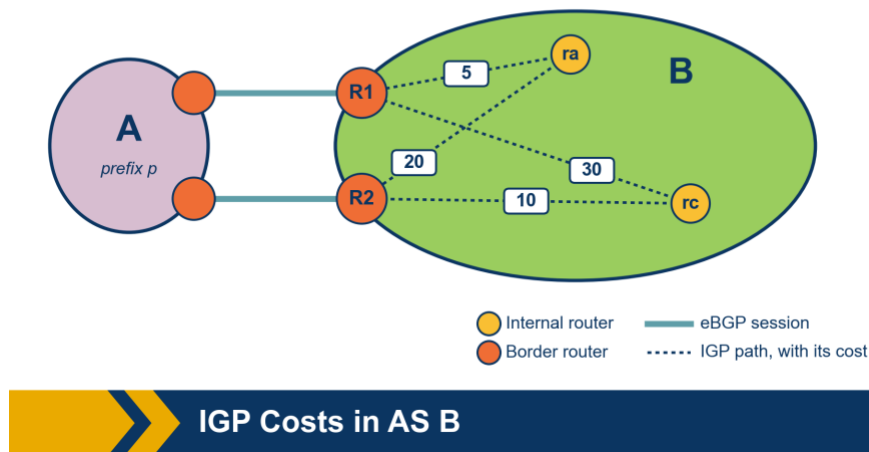


Figure: IGP costs in AS B (Module 4)

Consider AS B. Border routers R1 and R2 of AS B connect to AS A. The routers of AS B run a full mesh of iBGP sessions. Border routers R1 and R2 both hold routes to prefix p, and no other border router of AS B holds one. The two routes tie on LocalPref, AS path, origin type and MED. Internal router ra reaches R1 at IGP cost 5 and R2 at IGP cost 20. Internal router rc reaches R1 at IGP cost 30 and R2 at IGP cost 10.

Through which border router does each internal router send its traffic to p?

- **A.** Both ra and rc exit at R2, which has the lower total IGP cost.
- **B.** ra and rc alternate between R1 and R2 to balance the traffic load.
- **C.** ra exits at R1 and rc exits at R2, each by its own lowest IGP cost.
- **D.** ra and rc both exit at the border router with the lower traffic load.

Correct answer: **C**

Why: The two routes tie on every attribute compared before IGP cost, and each internal router picks the border router that is cheapest for it to reach. For ra, R1 costs 5 and R2 costs 20, and ra sends its traffic through R1. For rc, R1 costs 30 and R2 costs 10, and rc sends its traffic through R2. This is how two routers in the same AS end up using different exits for the same prefix. The decision process does not add the costs of different routers, and it does not look at traffic load. It picks one route for each router and does not switch between exits.

Q29. [MCQ]

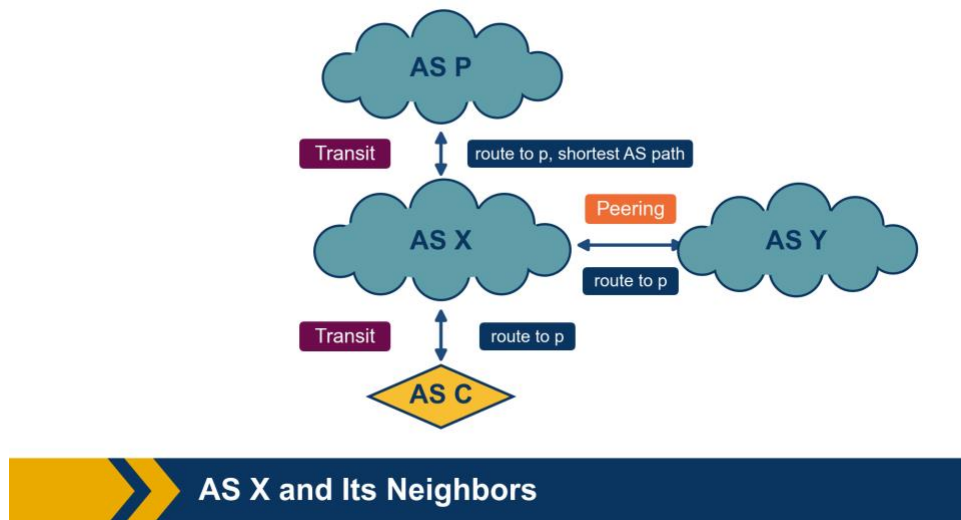


Figure: AS X and its neighbors (Module 4)

Consider AS X and its neighbors. AS C is a customer of AS X. AS Y is a peer of AS X. AS P is a provider of AS X. Each of the three neighbors advertises a route to prefix p to AS X. The route from AS P has the shortest AS path of the three. AS X sets LocalPref according to its business relationship with each neighbor.

Which LocalPref values does AS X assign to the three routes?

- A. Customer C 95, peer Y 85, provider P 75
- B. Customer C 75, peer Y 85, provider P 95
- C. Customer C 95, peer Y 75, provider P 85
- D. Customer C 85, peer Y 95, provider P 75

Correct answer: **A**

Why: AS X prefers the route with the highest LocalPref. Traffic sent to a customer earns AS X money, traffic sent to a peer is usually free, and traffic sent to a provider costs AS X money. The customer route gets the highest value, the peer route the middle value, and the provider route the lowest. Because LocalPref is compared before AS path length, AS X uses the customer route even though the route from the provider is shorter.

Challenges with BGP: Scalability and Misconfigurations

Q30. [MCQ]

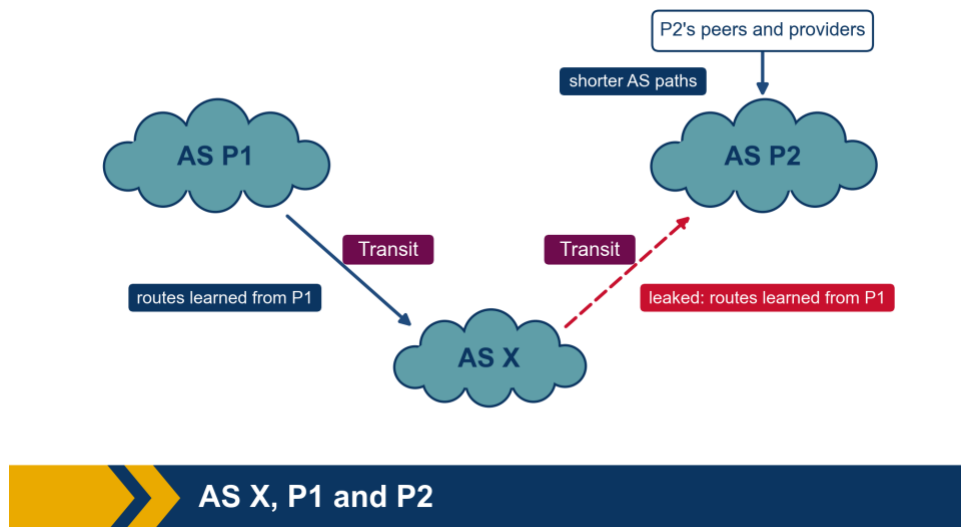


Figure: AS X, P1 and P2 (Module 4)

Consider AS X, a small AS with two providers, P1 and P2. AS X misconfigures its export policy and advertises to P2 all the routes that it learns from P1. P2 does not filter the leaked routes. P2 treats the leaked routes as customer routes. P2 has also learned routes to the same prefixes from its peers and providers. Many of the leaked routes have longer AS paths than these peer and provider routes.

What is the result of this misconfiguration?

- A. P2 ignores the leaked routes, since their AS paths are longer.
- B. P2 sends traffic for those prefixes to X, which X must carry to P1.
- C. P2 rejects the leaked routes, since BGP checks who owns each prefix.
- D. P2 keeps the leaked routes as backups, below its peer routes.

Correct answer: **B**

Why: P2 treats the leaked routes as customer routes and gives them its highest LocalPref. LocalPref is compared before AS path length, so the leaked routes win despite their longer AS paths, and P2 does not hold them as backups. P2 then sends its traffic for those prefixes to AS X, which must pay P1 to carry it and earns nothing from it. BGP has no check on prefix ownership that could stop the leaked routes.

Q31. [TF]

AS 64500 owns prefix 203.0.113.0/24 and originates it. AS 64511 has no right to this prefix. AS 64511 starts to announce prefix 203.0.113.0/24 to its BGP neighbors.

The neighbors of AS 64511 can tell from the BGP messages alone that AS 64511 does not own the prefix.

- True
- False

Correct answer: **False**

Why: BGP has no way to verify which AS owns a prefix, and any BGP router can announce a route for any prefix. The neighbors of AS 64511 therefore cannot tell from the BGP messages that AS 64511 does not own 203.0.113.0/24. The same gap let the false route

spread in the 2008 YouTube incident, where the upstream provider did not check the announcement against the prefixes its customer owned.

Q32. [TF]

A router has learned a route to prefix 198.51.100.0/22. The router then learns a route to prefix 198.51.100.0/24 from a different neighbor. Both routes are now in the forwarding table of the router.

For a packet sent to 198.51.100.7, the router uses the route to 198.51.100.0/24.

- True
- False

Correct answer: **True**

Why: The address 198.51.100.7 matches both prefixes. When an address matches several prefixes, the router uses the most specific one, the longest prefix. The /24 prefix is more specific than the /22 prefix, so the router uses the route to 198.51.100.0/24.

Peering at IXPs

Q33. [MCQ]

Consider an IXP. Several ASes are members of the IXP. Each member connects its border router to a port at the IXP. The members exchange traffic with each other through the IXP.

How does the IXP handle the traffic between its members?

- A. It carries the traffic as a transit provider and charges by volume.
- B. It passes the traffic through its route server, along with the routes.
- C. It routes each IP packet at layer 3, choosing the best path between the members.
- D. It switches layer 2 frames, and the members' routers make the routing decisions.

Correct answer: **D**

Why: An IXP works at layer 2. Its switching fabric forwards frames from the router of one member to the router of another, and the routers of the members make all the routing decisions. The route server handles routing information only, and the data packets cross the switching fabric directly. The IXP is not a transit provider, and it typically does not charge for the volume of traffic exchanged.

Q34. [TF]

Consider two ISPs that peer publicly at an IXP. The two ISPs exchange a large volume of traffic over the public peering link.

The IXP charges each ISP for the volume of traffic that the two ISPs exchange.

- True
- False

Correct answer: **False**

Why: IXPs typically do not charge for the volume of traffic exchanged over a public peering link. Each ISP pays for its circuit to the IXP, a monthly charge for its port, and possibly a yearly membership fee.

Peering at IXPs: How Does a Route Server Work?

Q35. [TF]

AS A and AS B are members of an IXP. AS A and AS B peer multi-laterally through the route server of the IXP. AS A and AS B do not have a separate bilateral BGP session. AS A sends data packets to a host in the network of AS B.

The data packets pass through the route server on their way to AS B.

- True
- False

Correct answer: **False**

Why: Only routing information passes through the route server. The data packets of AS A cross the IXP's switching fabric directly to the router of AS B.

Q36. [MCQ]

An IXP has 400 members. Of these members, 250 use the route server of the IXP. AS M is one of the 250 route server users. AS M peers multi-laterally through the route server and has no bilateral BGP sessions.

How many BGP sessions does AS M maintain at the IXP?

- A. 1
- B. 0
- C. 399
- D. 249

Correct answer: **A**

Why: AS M needs a single BGP session, the one to the route server. Through it, AS M peers with every other member that also uses the route server, without a separate session to each of the other 249 users. AS M has no session with the IXP switch, and none with the 150 members that do not use the route server.

Q37. [MCQ]

AS X is a member of an IXP. AS X announces prefix p to the route server of the IXP. The route server builds its import filters from a route registry. The route registry does not list p as a prefix that AS X may advertise. AS Z is another member of the IXP and also uses the route server.

What happens to the announcement of prefix p?

- **A.** The import filter drops p, so p never reaches the master RIB.
- **B.** The route server advertises p, but blocks data sent toward p.
- **C.** p enters the master RIB, and an export filter keeps it from Z.
- **D.** The route server advertises p to Z, since X's export filter allows it.

Correct answer: **A**

Why: The route server places p in the RIB of AS X and applies the import filter for that RIB, which checks whether AS X may advertise p. That filter is built from the route registry, and the registry does not list p for AS X. The filter drops p, so p never reaches the master RIB and no member receives it. Export filters act only on routes that pass the import filter. The route server handles routes, not data packets, and has no traffic to block.

Q38. [MCQ]

AS X, AS Y and AS Z are members of an IXP. All three use the route server of the IXP. The route server keeps a separate routing table (RIB) for each member.

AS X and AS Y both advertise a route to prefix p to the route server. The route from AS X has a shorter AS path than the route from AS Y. AS X has set an export filter so that the route server does not pass the routes of AS X to AS Z.

What does AS Z receive from the route server for prefix p?

- **A.** Both routes, so that the router of AS Z can run the best path selection process.
- **B.** No route, because the route from AS X, which has the shorter AS path, is blocked for AS Z.
- **C.** The route from AS X, because export filters apply only to bilateral peers.
- **D.** The route from AS Y, because no filter blocks that advertisement.

Correct answer: **D**

Why: The route server keeps a separate RIB for each member and picks the best route for each member separately. The export filter of AS X keeps the route from AS X away from AS Z. For AS Z, the only route to p is the route from AS Y, so the route server sends that route to AS Z. AS Z would receive no route only if the route server picked one best route for all members. That failure is called the hidden path problem.

Q39. [MCQ]

AS C, a content provider, is a member of a large IXP which runs a route server. Most of the traffic of AS C is exchanged with five other members of the IXP. AS C wants to use MED and AS path prepending to steer this traffic.

How should AS C set up its peering at the IXP?

- **A.** Peer with every member, including the five, only through the route server.
- **B.** Peer bilaterally with every member, and stop using the route server.
- **C.** Peer with the five through the route server, and bilaterally with the rest.

- **D.** Peer bilaterally with the five, and through the route server with the rest.

Correct answer: **D**

Why: A route server does not support AS path prepending, MED or scoped advertisements. AS C needs bilateral sessions with the five members to steer its traffic with these attributes. Peering with the five through the route server would leave AS C unable to steer most of its traffic. Members commonly do both, with bilateral sessions to their most important peers and the route server for everyone else. If a bilateral session goes down, the route server session serves as a backup.